Stroke Risk Prediction Proposal

ITAI 1371 Machine Learning Midterm Project • ***Team:*** *Karima, Danh, Great, Quan, Camila*

> **Original Kaggle Source:** Healthcare Disease Prediction Dataset
> **Working Copy:** Shared as a Google Sheet (link in dataset_url.txt).

1. Dataset

Patient health records used to predict whether a patient has had a stroke.

Property	Detail
Size	2,000 rows × 15 columns (14 after removing the `Patient_ID` identifier)
Target	Stroke (0 = no, 1 = yes); about 5% positive, heavily imbalanced
Numeric	Age, BMI, Systolic/Diastolic blood pressure, Cholesterol, Glucose level
Categorical	Gender; Smoking, Alcohol, Physical activity, Family history, Heart disease, Diabetes

2. Problem We Want to Solve

Stroke is a leading cause of death and disability. Early identification allows earlier intervention. We will build a binary classifier predicting stroke from routine clinical measurements and lifestyle factors, prioritizing high recall to minimize missed cases.

3. Midterm Accomplishments

- Split data 70% train / 30% test in Python; run EDA exclusively on the training set.
- Clean data: handle missing values, one-hot encode Gender, engineer clinical features, scale, and balance with SMOTE-NC.
- Visualize distributions via Matplotlib plots and inspect printed samples in a Jupyter notebook.

4. Final Project Plan

- Compare Logistic Regression, Random Forest, Gradient Boosting, and an additional model using stratified cross-validation.
- Evaluate via Precision, Recall, F1, PR-AUC, and ROC-AUC; tune decision thresholds.
- Interpret feature importance to confirm clinical validity (age, blood pressure, smoking).

5. Risks and Limitations

- Severe class imbalance (~5% positive) makes accuracy misleading with few positive test cases.
- Synthetic data risk: if simple rules dictate targets, high scores may reflect generation artifacts rather than real-world utility (checked in notebook).